

COLD CLIMATE HEAT PUMP OUTDOOR UNIT SCHEDULE - HP0U

	UNIT ID	SERVICE	LOCATION	UNIT TOTAL VOLT (VOLT)	UNIT ELEC. PHASE (WCA)	MAX. OVER-CURRENT (A)	COOLING TOTAL SEER (BTU/HM)	COOLING SEER (BTU/HM)	HEATING COP @ 47° F OAT	HEATING COP @ 5° F OAT	HEATING MAX. CAP. @ -2° BTU/W (MBT/h)	HSPF (SEER x 3.41) (FT)	APPROX. LINE LENGTH (FT)	L.G. LINE (IN)	GAS LINE (IN)	MANUF. OR EQUAL	MODEL OR EQUAL	REMARKS
HP0U-1	FIRST FLOOR	SOUTH ELEV	240/1	42	50	39.0	20	0.83	3.8	2.0	38.2	11.5	42	3/8	5/8	MITSUBISHI	MZS4MZ2MAH2U1T	BRANCH BOX: BB-1 PAC-KAKA23C
HP0U-2	SECOND FLOOR	SOUTH ELEV	240/1	42	50	33.8	20.7	0.92	3.8	2.0	39.8	12.1	72	3/8	5/8	MITSUBISHI	MZS4MZ2MAH2U1T	BRANCH BOX: BB-2 PAC-KAKA23C

COOLING CAPACITY RATINGS AT 95 °F OAT, 80/67 °F RETURN AIR, AS PER ARIH 210/240. UTILITY INSULATED LINE SETS EBRILLE INDUSTRIES USA. COVERALL OR EQUAL.

ALL CONDENSATE DRAINS: ¾" SCH40 PVC PIPED TO INDIRECT DRAINS FURNISHED BY PLUMBING SUBCONTRACTOR

COLD CLIMATE HEAT PUMP INDOOR UNIT SCHEDULE - HP1U

UNIT ID	LOCATION	SERVICE	AIR FLOW RATE (CFM)	EXT. STATIC FAN SETTING (IN WG)	FAN SPEED	UNIT ELEC. PHASE	UNIT ELEC. CAP. (kW)	ADJ. HEATING CAP. (kW)	TOTAL COOLING CAP. (kW)	L.I.D. LINE LENGTH (IN)	GAS APPROX. LINE LENGTH (FT)	MANUF. OR EQUAL	MODEL OR EQUAL	CONTROLS	REMARKS / ACCESSORIES	
HP1U-1	MECH ROOM	FIRST FLOOR GARHEATING, EXHAUST, ENTRY	760	0.5	MED.	240/1	4.2	25.4	27.0	3/8	5/8	24	MITSUBISHI	SUZAKP30NA	PAC-USWSH002-2, MHK2	HUMIDIFIER: H-1, APRILAIRE 600, OR EQUAL.
HP1U-2	BASEMENT	FIRST FLOOR BEDROOM SUITE	435	0.5	MED.	240/1	3.0	12.7	12.0	1/4	3/8	10	MITSUBISHI	SUZAKP12NA	PAC-USWSH002-2, MHK2	DUCTED DEHUMIDIFIER: DU-1; SEE REHEAT COILS
HP1U-3	ATTIC	BEDROOM 1 SUITE	275	0.2		240/1	1.0	10.0	8.1	1/4	3/8	12	MITSUBISHI	SEZAD09N4AR1.1H	PAC-USWSH002-2, MHK2	HUMIDIFIER: H-2, BELOW REHEAT COILS
HP1U-4	ATTIC	BEDROOM 2 SUITE	195	0.2		240/1	1.0	10.0	8.1	1/4	3/8	12	MITSUBISHI	SEZAD09N4AR1.1H	PAC-USWSH002-2, MHK2	
HP1U-5	ATTIC	GUEST BEDROOM	180			240/1	1.0	6.8	6.0	1/4	3/8	16	MITSUBISHI	MLZKVO8NA-U1	PAC-USWSH002-2, MHK2	
HP1U-6	ATTIC	BATHROOM, ENTRY, LANDING	355	0.2		240/1	1.0	12.8	11.4	1/4	3/8	12	MITSUBISHI	SEZAD012NAAR1.1H	PAC-USWSH002-2, MHK2	

*ADJUSTED CAPACITY RATINGS BASED ON INSET LENGTHS AND CLIMATE AS PER MITSUBISHI DIAMOND DESIGNER - SEE DRAWING H4

SUPPLEMENTAL HEATING COIL - RHC

UNIT ID	LOCATION	SERVICE	AIR FLOW RATE (CFM)	MAX. DISCHG AIR TEMP (°F)	MAX. ELEC DROPPHASE (IN. WG)	UNIT PRESS. (VOLT)	CURRENTPHASE (A)	CAPAC. (kW)	MCA (A)	MANUF. OR EQUAL	MODEL OR EQUAL	REMARKS
RHC-1	HP0U-1	SUPPLY AIR	760	200	0.05	240/1	33.3	8	41	MITSUBISHI	EH08-SV2-M	SEPARATE POWER KIT (SPRT); AUX HEAT LOCKOUT (ETC-21-20-MMT)
RHC-2	HP0U-2	SUPPLY AIR	435	200	0.05	240/1	20.8	5	28	MITSUBISHI	EH05-SV2-S	SEPARATE POWER KIT (SPRT); AUX HEAT LOCKOUT (ETC-21-20-MMT)

INSTALLATION DETAILS: http://myfieldnote.com/USA/APP_Notes/Application_Note_1030_ME-SV2Z_and_PVA_Electric_Heater_Configuration_011620

ENERGY RECOVERY VENTILATOR SCHEDULE - ERV

UNIT ID	LOCATION	SERVICE	MIN. FLOW RATE (CFM)	EXTERNAL STATIC PRESS. (IN. WG)	UNIT ELEC. PHASE	UNIT ELEC. CAP. (MCA)	EXP. POWER (W)	EFFEC. @ 32° F OAT (%)	MANUF. OR EQUAL	MODEL OR EQUAL	WT. (LBS)	REMARKS
ERV-1	MECH ROOM 003	ALL AREAS	220	0.8	115/1	10	185	61	RENEWAIRE	PREMIUM L	52	PBL BOOST TIMERS (3)

DEHUMIDIFICATION UNIT - DU

UNIT ID	LOCATION	SERVICE	AIR FLOW RATE (CFM)	EXTERNAL STATIC PRESS. (IN. WG)	UNIT ELEC. PHASE	UNIT ELEC. (MCA)	CAP. (PINTS/DAY)	ENERGY FACTOR (L/kWh)	MANUF. OR EQUAL	MODEL OR EQUAL	WT. (LBS)	REMARKS
DU-1	MECH ROOM 003	FIRST FLOOR ZONE 1	270	0.2	115/1	8.3	105.0	2.35	APRILAIRE	ET130	98	SUSPEND FROM CEILING. SEE H-2 DETAIL.

CAPACITY BASED ON INLET CONDITIONS: 73 °F DBT, 60% RH

ADD ALTERNATE: MAKE UP AIR UNIT - MUA

UNIT ID	LOCATION	SERVICE	MAX. FLOW RATE (CFM)	EXTERNAL STATIC PRESS. (IN. WG)	UNIT ELEC. PHASE	UNIT ELEC. (MCA)	FAN ELEC. POWER (kW)	ELEC. RESIST. OR EQUAL	MANUF. OR EQUAL	MODEL OR EQUAL	WT. (LBS)	REMARKS
MUA-1	MECH ROOM 008	KITCHEN/PANTRY	600	0.4	115/1	1.4	90	6	FANTECH	MUAS 750	121	VARIABLE FLOW RATE BASED ON KEF SPEED. SEE WIRING DIAGRAM.

KITCHEN EXHAUST FAN (KEF) FURNISHED BY OWNERS. COORDINATE WITH ELECTRICIAN FOR MUA CONTROLLER INTERFACE

DIFUSER, REGISTER AND GRILLE SCHEDULE

ID	LISTED SIZE (WIDTH x HEIGHT)	AIR FLOW RATE (CFM)	THROW @ 7' (FT)	FACE VELOCITY (FPM)	SERVICE	MOUNTING	ACCESSORIES	MANUF. / MODEL, OR EQUAL	REMARKS
SD-A	10" x 4"	75 - 120	9.5 - 13	500 - 700	SUPPLY AIR	SIDEWALL	VOLUME DAMPER	HART AND COOLEY / 618	STEEL, PAINTED
SD-B	10" x 6"	121 - 190	9 - 14	500 - 700	SUPPLY AIR	SIDEWALL	VOLUME DAMPER	HART AND COOLEY / 618	STEEL, PAINTED
SD-C	10" x 8"	191 - 280	12 - 19	500 - 700	SUPPLY AIR	SIDEWALL	VOLUME DAMPER	HART AND COOLEY / 618	STEEL, PAINTED
SD-E	14" x 16"	501 - 600	13 - 20	500 - 700	SUPPLY AIR	SIDEWALL	VOLUME DAMPER	HART AND COOLEY / 618	ALUMINUM, PAINTED
SD-4	4" R	x <= 40			SUPPLY AIR	CEILING		FANTECH MGS4	STEEL, WHITE POWDER-COATED
SD-A	6" R	41 < x <= 85	3.5 - 4	< 700	SUPPLY AIR	CEILING	VOLUME DAMPER	HART AND COOLEY / 16	STEEL, PAINTED
SD-B	8" R	96 < x <= 160	5	<= 700	SUPPLY AIR	CEILING	VOLUME DAMPER	HART AND COOLEY / 16	STEEL, PAINTED
SD-C	10" R	161 < x <= 240	5	<= 700	SUPPLY AIR	CEILING	VOLUME DAMPER	HART AND COOLEY / 16	STEEL, PAINTED
SD-E-2	4" x 12"	x <= 90	10	300 - 400	SUPPLY AIR	FLOOR	VOLUME DAMPER	VENTCOVERSUNLIMITED.COM	
V-A	4" R	< 50			VENTILATION SYSTEM SUPPLY AND RETURN	SIDEWALL OR CEILING		SEHO / TT 4	SEE OWNER BEFORE PURCHASING
V-B	6" R	50 <= x <= 90			VENTILATION SYSTEM SUPPLY AND RETURN	SIDEWALL OR CEILING		SEHO / TT 6	SEE OWNER BEFORE PURCHASING
RG-1	12" x 8"	< 300			RETURN AIR	SIDEWALL OR CEILING			
RG-2	12" x 18"	< 450			RETURN AIR	SIDEWALL OR CEILING			
RG-3	14" x 18"	< 600			RETURN AIR	SIDEWALL OR CEILING			
RG-4	20" x 24"	< 800			RETURN AIR	SIDEWALL OR CEILING			
RG-5	24" x 16"	< 900			RETURN AIR	SIDEWALL OR CEILING			

*COORDINATE SELECTION OF ALL SELECTIONS OF DIFUSERS, REGISTERS AND GRILLES WITH THE OWNER.

INTEGRAL VOLUME DAMPER IF DUCT VOLUME DAMPER WILL NOT BE ACCESSIBLE

ELECTRIC FINNED TUBE RADIATION - EFTR

UNIT ID	SERVICE	LENGTH (FT)	DENSITY (W/ FT)	UNIT ELEC PHASE	CAPACITY (KW)	CAPACITY (MBR)	MANUF. OR EQUAL	MODEL OR EQUAL	REMARKS
EFTR-4	SUPPLEMENTAL HEATING	4	250	240/1	1.00	3.4	STELPRO	BRAVA AB41002	INTEGRAL DIGITAL THERMOSTAT
EFTR-5	SUPPLEMENTAL HEATING	5	250	240/1	1.25	4.3	STELPRO	BRAVA AB41252	INTEGRAL DIGITAL THERMOSTAT
EFTR-6	SUPPLEMENTAL HEATING	6	250	240/1	1.50	5.2	STELPRO	BRAVA AB41502	INTEGRAL DIGITAL THERMOSTAT
EFTR-8	SUPPLEMENTAL HEATING	8	250	240/1	2.00	6.8	STELPRO	BRAVA AB42002	INTEGRAL DIGITAL THERMOSTAT

COORDINATE COLOR WITH OWNER.

SYSTEMS DESCRIPTIONS

INSIDE DESIGN CONDITIONS			
SPACES	HEATING DBT (DEG. F)	COOLING DBT (DEG. F)	NOTES
BASEMENT, GARAGE	50	65	SPACE CONDITIONING INC.
FIRST FLOOR	72	74	
SECOND FLOOR	72	74	

OUTSIDE DESIGN CONDITIONS			
HEATING DBT (DEG. F)	COOLING DBT (DEG. F)	WBT (DEG. F)	NOTES
3 °F	87 °F	71 °F	BASED ON AUSTRALIA, ME 98% ASHRAE

CONDITIONED AREA (SA)
THE CONDITIONED AREAS IS THE FLOOR AREA (SA) WITHIN THE INSULATED BOUNDARY. THE LIST OF SPACES IS SUMMARIZED IN THE TABLE BELOW AND PROVIDED IN ATTACHMENT 1. EACH SPACE'S AREA IS A ROUNDED TO A MULTIPLE OF 5 SF. THE ATTIC IS NOT INCLUDED.

THE OUTDOOR AIR INfiltrATION RATE IS ESTIMATED TO BE 0.19 NATURAL AIR CHANGES PER HOUR, WHICH IS ACCA MANUAL J5 "SEMI-TIGHT" CONSTRUCTION FOR A RESIDENCE OF THIS SIZE IN AREA SQUARE FEET (SA). THIS ESTIMATE IS FOR THE PURPOSE OF LOAD CALCULATIONS; THE LIKELY LEAKAGE RATE FOR A CLOSED-CELL SPRAY FOAM BUILDING IS LESS THAN THE ESTIMATE.

VENTILATION RATE

VENTILATION (INCOMING FROM OUTDOORS) OR EXHAUST AIR OUTGOING AIR TO OUTDOORS MUST FLOW THROUGH ALL SPACES CONSTANTLY TO MAINTAIN GOOD AIR QUALITY. THE MINIMUM VENTILATION RATE, VIA STANDARD ASHRAE 62.2-2006, IS CALCULATED AS 0.03 CFM PER SQUARE FOOT TIMES THE CONDITIONED AREA PLUS THE NUMBER OF BEDROOMS PLUS 1 TIMES 7.5, OR, 0.03 X 5,910 + 7.5 X 6 = 215 CFM. ENERGY IS RECOVERED TO THE INCOMING AIR FROM THE EXHAUSTED AIR. THE STANDARD'S "INfiltrATION CREDIT" IS NOT ASSUMED. THE GOAL IS FOR ALL SPACES TO BE EITHER SUPPLIED OR EXHAUSTED TO AVOID MOISTURE AND ODOOR ACCUMULATION, AIR FLOW DESIGN CONCEPT.

IN GENERAL, EACH SPACE'S HEATING AND COOLING LOADS WILL BE SERVED BY A PROPORTIONATE VOLUME FLOW RATE OF CONDITIONED AIR. AIR WILL RETURN TO ITS SOURCE THROUGH COMMON SPACES AND AIR RETURN GRILLES.

VENTILATION AIR WILL BE SUPPLIED TO THE PERIMETER OF OCCUPIED AREAS AND RETURNED FROM BATHROOMS, CLOSETS, AND OTHER NORMALLY OR INTERMITTENTLY UNOCCUPIED SPACES. THE LOCATION OF THE SUPPLY AND RETURN OUTLETS IS INTENDED TO MAXIMIZE AIR FLOW COVERAGE.

HEATING, VENTILATION AND AIR CONDITIONING SYSTEMS DESCRIPTION

HEATING, AIR CONDITIONING SYSTEMS - CENTRAL HEAT PUMP AIR SYSTEMS
THE FIRST AND SECOND FLOORS PRIMARILY HEATING AND COOLING WILL BE COULD CLIMATE HEAT PUMPS WITH INDOOR AIR HANDLING (FIRST FLOOR). EACH INDOOR UNIT SERVES A HEATING AND COOLING ZONE. A ZONE IS ONE OR MORE SPACES THAT ARE SERVED AS A GROUP. SEE DRAWING H-4 FOR A HEAT PUMP SYSTEMS SCHEMATIC. EACH SPACE IS A NAMED AREA WITHIN THE BUILDING. SEE ATTACHMENT 1 FOR A LIST OF SPACES AND THEIR LOADS.

FIRST FLOOR

THE FIRST FLOOR WILL BE TWO ZONES. THE PRIMARY BEDROOM SUITE (ZONE 1) AND ALL OTHER SPACES (ZONE 2). EACH ZONE WILL BE SERVED BY AN AIR-HANDLING UNIT (AHU). THE AHU WILL BE SERVED BY ONE OUTDOOR UNIT VIA A BRANCH BOX. (PAC-KAKI HARDWARE INTERFACE. EACH AHU WILL HAVE AN ELECTRIC REHEAT COIL TO SERVE AS BACK-UP HEAT AND FOR HEATING ROOMS WHEN THE OUTDOOR UNIT DEFROSTS. EACH WILL HAVE A HUMIDIFIER. SEE DRAWING H-1 FOR THE LIST OF MAJOR EQUIPMENT.

EACH ZONE WILL BE CONTROLLED BY A WALL MOUNTED THERMOSTAT (MHK2) OR A SMARTPHONE VIA MITSUBISHI'S "MIMO CLIMATE" SMARTPHONE APP. THE USE OF THE APP IN CONJUNCTION WITH THE WALL THERMOSTAT IS THE MOST FLEXIBLE CONTROL APPROACH. THE WALL THERMOSTAT SERVES AS THE LOCATION THAT TEMPERATURE IS MEASURED, IS PROGRAMMABLE, AND HAS A TYPICAL THERMOSTAT'S TOUCH-SCREEN INTERFACE. IT WILL BE CONVENIENT FOR OCCUPANTS WITHOUT APP ACCESS. THE APP IS AN INTERFACE TO MITSUBISHI'S SERVERS VIA THE SITE'S WIFI SYSTEM, AND IS THE MOST CONVENIENT FOR ACCESS, CONTROL, AND PROGRAMMING, LOCALLY OR REMOTE FROM THE RESIDENCE.

THE FIRST FLOOR'S AIR HANDLING UNITS AND THEIR BRANCH BOX WILL BE IN THE BASEMENT. AN AIR DUCT DISTRIBUTION SYSTEM WILL BE INSTALLED AND TO SERVE EACH SPACE. SEE DRAWINGS H-2 AND H-3.

SECOND FLOOR

THE SECOND FLOOR WILL BE ZONED INTO FOUR AREAS, EACH SERVED WITH A FAN-COIL UNIT (SE-K10 OR A CASSETTE (MLZ-KV)). THE INDOOR UNITS WILL BE SERVED FROM ONE OUTDOOR UNIT VIA A BRANCH BOX HARDWARE INTERFACE. THE FAN-COIL UNITS AND BRANCH BOX WILL BE IN THE ATTIC. SERVICE ACCESS IS NEEDED FOR THESE UNITS. SEE DRAWINGS H-4 AND H-5.

THE FAN-COIL UNITS AND THEIR BRANCH BOX WILL BE IN A CENTRAL ATTIC AREA ABOVE THE CLOSETS AND HALL SPACES. THE CASSETTE WILL BE SURFACE MOUNTED IN THE GUEST BEDROOM.

EACH ZONE WILL BE CONTROLLED BY A WALL MOUNTED THERMOSTAT (MHK2) OR THE KIMBO APP.

OUTDOOR UNITS

OUTDOOR UNITS (MAX 5M) WILL BE LOCATED AT GRADE ON THE SOUTHEAST SIDE OF SPACE 113. THE FIRST FLOOR UNITS' NOMINAL COOLING CAPACITY IS 3.5 TONS; THE SECOND FLOOR UNITS' NOMINAL CAPACITY IS 3.0 TONS. A SHED ROOF ABOVE THE UNITS WILL BE CONSTRUCTED TO SHIELD THEM FROM ROOF RAINFALL.

VENTILATION SYSTEM APPLIANCE

AN ENERGY RECOVERY VENTILATOR (ERV) WILL MAINTAIN INDOOR AIR QUALITY BY REPLACING INDOOR AIR WITH OUTDOOR, AND RECOVERING ENERGY TO FLOW INDOOR AIR STREAM TO THE INCOMING AIR. THE DESIGN FLOW RATE OF BOTH AIR STREAMS IS 220 CFM. "BALANCED" VENTILATION, AIR DISCHARGING FROM THE ERV TO SPACES IS ITS "SUPPLY"; AIR RETURNING TO THE ERV FROM SPACES IS ITS "RETURN". OUTDOOR AIR TO THE ERV IS ITS OUTSIDE AIR "INTAKE". AIR DISCHARGED FROM THE ERV TO OUTDOORS IS ITS "EXHAUST". THE ERV WILL BE PLACED IN THE ATTIC. THE INCOMING AIR STREAM IS FILTERED. REMOVAL SERVICE WILL BE NEEDED FOR THIS UNIT FOR FILTERS, DEFROSTING AND CLEANING.

THE ERV'S SHOULD OPERATE AT MINIMUM WHENEVER THE RESIDENCE IS OCCUPIED AND ITS WINDOWS ARE CLOSED. THESE UNITS SERVE AS BATHROOM FOR EXHAUST VIA A LOCAL "BOOST" SWITCH-DEDICATED BATHROOM FANS ARE NOT NEEDED.

DEHUMIDIFICATION APPLIANCE
A DEHUMIDIFICATION SYSTEM, DU-1, WILL BE INSTALLED IN CONJUNCTION WITH HP1U-1 TO PROVIDE A "DEHUMIDIFICATION MODE". ITS MAIN GOAL IS TO REMOVE MOISTURE FROM THE AIR DURING PERIODS THAT HP1U-1 IS NOT OPERATING BECAUSE THE ROOM TEMPERATURE SET POINT IS REACHED.

AT A DEW POINT TEMPERATURE OF 55 DEG. F THE ENVIRONMENT IS "DRY AND COMFORTABLE". BETWEEN 55 DEG. F AND 65 DEG. F, THE ENVIRONMENT IS "INCENSESIVELY "STICNY", GREATER THAN 65 DEG. F, IT'S "MUGGY". DU-1'S SELECTION IS BASED ON A "DEHUMIDIFICATION DESIGN-DAY". THE OUTSIDE AIR TEMPERATURE IS 73 DEG. F, THE DEW POINT TEMPERATURE IS 71 DEG. F.; THE RELATIVE HUMIDITY IS 91%. AT THESE CONDITIONS, APPROXIMATELY 4.6 POUNDS OF WATER WILL BE REMOVED AND THE RESULTING AIRS DEW POINT TEMPERATURE WILL BE REDUCED TO 55 DEG. F

THE DEHUMIDIFICATION MODE REQUIRES THAT HP1U-1'S SUPPLY FAN OPERATE WHILE DU-1 IS OPERATING.
KITCHEN RANGE HOOD EXHAUST AND MAKE-UP AIR SYSTEM
A FAN AND DUCT SYSTEM TO SUPPLY REPLACEMENT AIR FOR THE KITCHEN HOOD'S EXHAUST AIR DURING ITS OPERATION IS RECOMMENDED FOR THIS RESIDENCE, BECAUSE ITS RANGE HOOD EXHAUST IS EXPECTED TO EXCEED 400 CFM. THE PROPOSED "MAKE-UP AIR SYSTEM, MUA-1, IS IN THE GARAGE. ITS CONTROL SYSTEM WILL SERVE THE FLOW RATE OF THE EXHAUST FAN AND SUPPLY AN EQUAL FLOW RATE OF MAKE-UP AIR.

ITS ACCESSORIES INCLUDE AN ELECTRIC COIL TO TEMPER THE MAKE-UP AIR TO 60 DEG. F (ADJUSTABLE). THE HEATER WILL HAVE ITS OWN DISCONNECT SWITCH. OTHER ACCESSORIES INCLUDE A FILTER SECTION AND FLOW STRAIGHTENERS.
CLOTHES DRYER EXHAUST
A 4" STEEL DUCT WILL BE PROVIDED FOR CLOTHES DRYER EXHAUST.
BACK-UP AND SUPPLEMENTAL ELECTRIC HEAT

ELECTRIC HEAT IS PROVIDED AS A BACK-UP SUPPLEMENTAL HEAT, AND UNOCCUPIED HEAT SOURCE. THE HEAT PUMP INDOOR UNITS SERVING THE FIRST FLOOR INCLUDE ELECTRIC HEATING COILS. ELECTRIC FINNED TUBE RADIATION IS PROVIDED FOR THE BEDROOMS. SHOULD THE HEAT PUMP SYSTEM NEED SERVICE, THE ELECTRIC HEAT CAN BE UTILIZED. DURING UNOCCUPIED PERIODS THE BEDROOMS COULD BE HEATED TO A SETBACK TEMPERATURE BY THE FINNED TUBE RADIATION. THE IN-FLOOR HYDRONIC SYSTEM HEATING ON THE BASEMENT WILL PROVIDE A SEPARATE WIDE SOURCE OF BUILDING HEAT.

HVAC LEGEND

CAP	CAPACITY	SAO	ERV - SUPPLY AIR OUT
CD	CEILING DIFUSER	SD	SUPPLY AIR DIFUSER - SEE SCHEDULE
CFM	CUBIC FEET PER MINUTE - AIR FLOW RATE	SEER	SEASONAL ENERGY EFFICIENCY RATIO
COP	COEFFICIENT OF PERFORMANCE	SG	SUPPLY AIR GRILLE WALL - SEE SCHEDULE
D	DEHUMIDISTAT	SR	SUPPLY AIR REGISTER - SEE SCHEDULE
DBT	DRY BULB TEMPERATURE	T	THERMOSTAT
EAO	ERV - EXHAUST AIR OUT	TV5	TERMINAL VELOCITY 75 FPM
EG	ERV EFFECTIVENESS	TV6	TYPICAL
FPM	FEET PER MINUTE	V	VENTILATION / HRV/ ERV - SUPPLY OR RETURN
HP1U	HEAT PUMP INDOOR UNIT	Vb	VENTILATION BOOST SWITCH
HP0U	HEAT PUMP OUTDOOR UNIT	Vc	VENTILATION CO2 CONTROL
HRV	HEAT RECOVERY VENTILATOR	Vd	VOLUME DAMPER
HSF2 (N)	HEATING SEASON PERFORMANCE FACTOR, VERSION 2, ASHRAE REGION 4	W	WATTS
		WBT	WET BULB TEMPERATURE

IN WIG
KIV
KWH
KILOWATT-HOUR
THOUSAND BTUS PER HOUR
MBH
THOUSAND BTUS PER HOUR
Ø
DIAMETER
OAI
OUTSIDE AIR IN
OAT
OUTSIDE AIR TEMPERATURE
RAI
RETURN AIR IN
RAI
RETURN AIR PATH DEVICE - SEE SCHEDULE
RHC
RETURN REGISTER - SEE SCHEDULE

GENERAL NOTES

THE DRAWINGS ARE DIAGRAMMATIC AND INDICATE GENERALLY THE LOCATION OF MATERIALS AND EQUIPMENT. THE CONTRACTOR SHALL FURNISH AND INSTALL ALL LABOR, EQUIPMENT, AND MATERIALS SHOWN AND REQUIRED TO HAVE A COMPLETE AND FUNCTIONING SYSTEM TOGETHER WITH ALL ASSOCIATED APPARATUS AS SHOWN ON THE PLANS. "PROVIDE" SHALL MEAN TO FURNISH AND INSTALL.

INFORMATION AND COMPONENTS SHOWN ON FLOW DIAGRAMS AND DETAIL DRAWINGS BUT NOT SHOWN ON FLOOR PLANS, AND VICE VERSA, SHALL APPLY AND SHALL BE PROVIDED AS THOUGH REQUIRED EXPRESSLY BY BOTH. IT IS NOT INTENDED TO SHOW EVERY OFFSET, FITTING OR COMPONENT; HOWEVER, CONTRACT DOCUMENTS REQUIRE COMPONENTS AND MATERIALS WHETHER OR NOT INDICATED OR SPECIFIED AS NECESSARY TO MAKE THE INSTALLATION COMPLETE AND OPERATIONAL.

FINAL LOCATION OF EQUIPMENT, AND CONNECTION POINTS SHALL BE DETERMINED IN THE FIELD WITH THE CONTRACTOR BEING RESPONSIBLE FOR DIMENSIONS THAT SHALL BE CONFIRMED AND CORRELATED AT THE JOB SITE.

THE WORK SHALL BE DONE IN ACCORDANCE WITH THE LOCAL BUILDING MECHANICAL CODES, NFPA AND ALL OTHER APPLICABLE STATE AND FEDERAL CODES.

INSTALL WORK SO AS TO BE READILY ACCESSIBLE FOR OPERATION, MAINTENANCE, AND REPAIR. MINOR DEVIATIONS FROM THE DRAWINGS MAY BE MADE TO ACCOMPLISH THESE GOALS.

ALL DUCT AND TERMINALS SHALL BE INSTALLED APPROXIMATELY AS INDICATED AND MODIFIED TO AVOID INTERFERENCE.

TO MAINTAIN CLEARANCE AND MINIMIZE PRESSURE DROP, DUCT FITTINGS AT THE AHU DUCT DISCHARGE, RETURN PLenums AND END-OF-DUCT DISTRIBUTION ARE INDICATED TO MINIMIZE PRESSURE DROP.

GUARANTEE WORK IN WRITING FOR ONE YEAR FROM DATE OF FINAL NOTICE OF ACCEPTANCE. REPAIR OR REPLACE DEFECTIVE MATERIALS, EQUIPMENT, WORKMANSHIP AND INSTALLATION THAT DEVELOP WITHIN THIS PERIOD. PROMPTLY AND TO THE OWNER'S SATISFACTION AND CORRECT DAMAGE CAUSED IN MAKING NECESSARY REPAIRS AND REPLACEMENTS UNDER GUARANTEE WITHIN CONTRACT.

ASSURE ADEQUATE SUPPORT FOR DUCT AND EQUIPMENT. PREVENT VIBRATION AND SWAYING.

PREVENT ENTRY OF FOREIGN MATTER DURING CONSTRUCTION.

PROGRESS

<i>General Contractor</i> RON DUNN BUILDERS Farmington, ME https://dunndbuilders.net 207-314-3911	
<i>HVAC Design</i> OCTOBER ENGINEERING LLC Palemto, ME; Sudbury, MA https://octoberengineering.com 207-649-0505; 508-561-7553	
<i>Owners</i> RICHARD & WENDY HAENSLER Chelmsford, MA 973-220-8069	
RESIDENCE PARKHILL POINT ROAD BELGRADE, ME 04917	

HVAC SCHEDULES